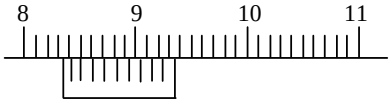
PHYSICS

SECTION – I
Single Correct Choice Type

This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

---

1. The length of a plate is measured with the help of a vernier calliper having zero correction – 0.02 cm. The observation is shown in figure. Then length of the plate from these observations is



(a) 8.30 cm (b) 8.34 cm (c) 8.32 cm (d) 8.36 cm

2. A particle is moving on x-axis according to the equation $x = \sin \omega t$. Consider the motion of the particle from $t = 0$ sec . to $t = 2\pi$ sec. Choose the correct statement.

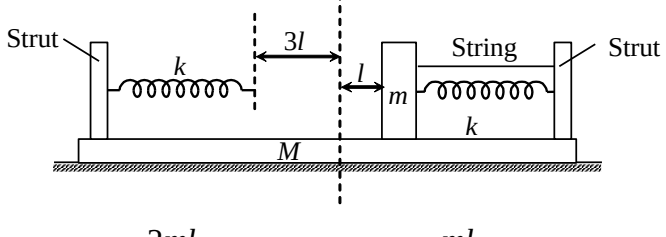
(a) Particle is retarding from $t = 0$ sec. to $t = \pi$ sec.

(b) Particle is retarding from $t = 0$ sec. to $t = \frac{\pi}{2}$ sec. and from $t = \pi$ sec. to $t = \frac{3\pi}{2}$ sec.

(c) Particle is retarding from $t = \frac{\pi}{2}$ sec. to $t = \frac{3\pi}{2}$ sec.

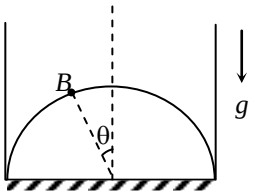
(d) Particle is retarding from $t = \frac{\pi}{2}$ sec. to $t = \pi$ sec. and from $t = \frac{3\pi}{2}$ sec. to $t = 2\pi$ sec.

3. A plank of mass M is placed on a smooth horizontal surface. Two light identical springs each of stiffness k are rigidly connected to struts at the ends of the plank as shown in the figure. When the springs are in their unextended position the distance between their free ends is $3l$. A block of mass m is placed on the plank and pressed against one of the springs so that it is compressed by l. To keep the block at rest it is connected to the strut by means of a light string. Initially the system is at rest. Now the string is burnt. Then the maximum displacement of the plank is



(a) $\frac{5ml}{m+M}$ (b) $\frac{2ml}{m+M}$ (c) $\frac{ml}{m+M}$ (d) $\frac{8ml}{m+M}$

4. A bead B of density ρ, which can slide on a friction less fixed semicircular structure of radius R is maintained at rest in an empty container as shown in figure. Now a liquid of density 2ρ is poured in the container upto height h ($h > R$) and the bead is released. Maximum contact force between bead and the structure at any time during the motion of the bead will be proportional to (θ is very small).



(a) $1 + \theta$ (b) $1 + \theta^2$ (c) $1 - \theta^2$ (d) $1 - \theta$

5. A monoatomic ideal gas undergoes a process $T^2 = 4P$. Molar specific heat of the process will be ($R \rightarrow$ universal gas constant)

(a) R (b) $\frac{5}{2}R$ (c) $\frac{R}{2}$ (d) $2R$

6. A block of mass M = 1 kg is dropped on a vertical spring, one end fixed to the ground. Height of the block from the free end of the spring is $\frac{5}{4}m$. Spring constant is $k = 5N/m$. After what time will the block come to its initial position again.

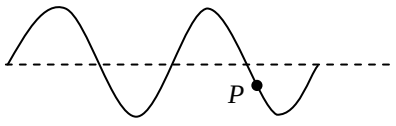
(a) $1+\frac{2\pi}{\sqrt{5}}+\frac{2}{\sqrt{5}}\sin^{-1}\left(\frac{2}{3}\right)$ (b) $1+\frac{2\pi}{\sqrt{5}}-\frac{2}{\sqrt{5}}\cos^{-1}\left(\frac{2}{3}\right)$

(c) $1+\frac{\pi}{\sqrt{5}}-\frac{2}{\sqrt{5}}\sin^{-1}\left(\frac{2}{3}\right)$ (d) $1+\frac{\pi}{\sqrt{5}}+\frac{2}{\sqrt{5}}\cos^{-1}\left(\frac{2}{3}\right)$

7. A point object is placed at the centre of curvature on the principal axis of a convex lens of focal length 10 cm. If a slab of some material of thickness 2 cm is introduced between the object and the lens magnification becomes half. Refractive index of material is (upto two significant digits)

(a) 0.16 (b) 0.17 (c) 0.18 (d) 0.19

8. A progressive wave heading in –ve *x* direction is maintained in a stretched string. At a certain instant snapshot of some part of string is taken which is as shown. Choose the correct direction of velocity of point *P* at this instant.



(a) upwards (b) downwards

(c) It may be upwards or downwards (d) Data insufficient

SECTION – II
Multiple Correct Choice Type

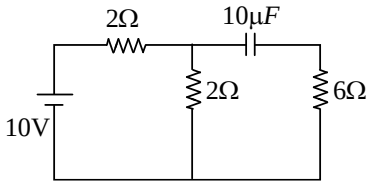
This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

---

9. Choose the correct option(s)

(a) Time constant of the circuit is 70 μsec

(b) In steady state, charge on the capacitor is 50 μC



(c) At any instant *t*, current through 6 Ω resistance is $\frac{5}{7}e^{-t/\tau}\ A\ (\tau = 70\,\mu\text{sec})$

(d) At any instant *t*, current through 6 Ω resistance is $\frac{3}{7}+\frac{5}{7}e^{-t/\tau}\ A\ (\tau = 70\,\mu\text{sec})$

10. A charge particle of specific charge $\left(\frac{q}{m}\right)$ 8 columb / kg is projected on a rough horizontal *X*-*Y* plane. Electric and magnetic fields are given by $\vec{E} = -5\hat{k}N/C$ and $\vec{B} = -10\hat{k}$ tesla respectively are present in the region. The particle enters in the region at (4, 0, 0) m with a velocity $100\,\hat{j}$ m/sec. If coefficient of friction between the particle and plane is $\mu = \frac{1}{3}$, then choose the correct option(s)

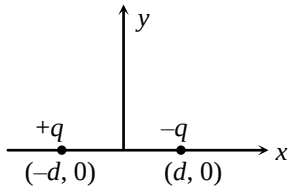
(a) The time after which particle comes at rest is 6 sec.

(b) Total work done by electric force on the particle is zero.

(c) Total distance covered by the particle is 300 m.

(d) Total work done by magnetic force on the particle is zero.

11. Two charges + q and – q are fixed on x-axis as shown. Consider a region in y-z plane $a^2 \le y^2 + z^2 \le b^2$. Choose the correct statement(s). ($a >>> d$)



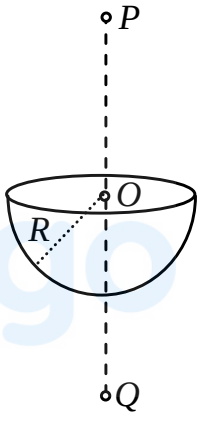
(a) Electric field anywhere in the given region is directed towards +ve x-axis.

(b) work done by the electric field in bringing a +ve test charge from $\left(0, \frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ to $\left(0, \frac{a}{\sqrt{2}}, \frac{-a}{\sqrt{2}}\right)$ is zero.

(c) Electric potential throughout the given region is zero.

(d) flux crossing this surface is $\frac{qd}{\in_0}\left(\frac{1}{a} - \frac{1}{b}\right)$

12. Consider a hemispherical body of uniform mass density ρ and radius R as shown. P and Q are two points such that $OP = OQ = 2R$ as shown. Choose the correct statement(s)



(a) Magnitude of gravitational field intensity at P and Q are same.

(b) Gravitational field intensity at P and Q are unlike parallel.

(c) Magnitude of gravitational field intensity at Q is $\frac{\pi G \rho R}{6}$

(d) If E_0 is the magnitude of gravitational field intensity at Q then at P magnitude of gravitational field intensity is $\frac{\pi G \rho R}{3} - E_0$.

SECTION–III
Comprehension Type

This section contains 1 group of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

A closed cylinder insulated from the surroundings is divided into two equal parts by means of an adiabatic piston. The left part is filled with 32 g of O_2 and 4 g of He and right part is vacuum. The initial temperature of the mixture is T_0. One third of oxygen gas dissociates into atoms. The piston is released and the gas fills the whole volume of the cylinder. The piston is slowly displaced back to its original position. Neglect further dissociation of gaseous molecules during motion of the piston.

(Take $2^{0.56} \approx \sqrt{2}$, R = 8.3 J/K/mol, $\sqrt{2} = 1.41$).

13. The adiabatic exponent for the mixture is

(a) 1.56 (b) 1.5

(c) 1.8 (d) 1.53

14. The temperature of the gas when piston returns back to its initial position is

(a) $T_0\sqrt{2}$ (b) $\frac{T_0}{\sqrt{2}}$

(c) $2T_0$ (d) $\frac{T_0}{2}$

15. The increase in internal energy of the gas is (in J)

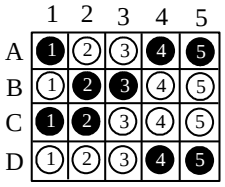
(a) 14.18 T_0 (b) 14 T_0

(c) 10 T_0 (d) zero

SECTION–IV
Matrix–Match Type

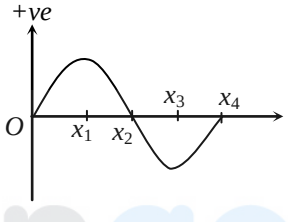
This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



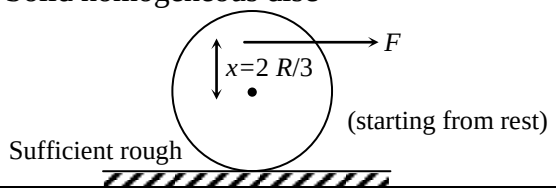
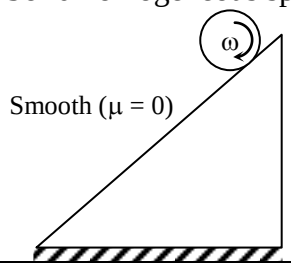
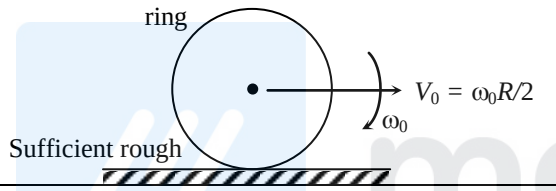
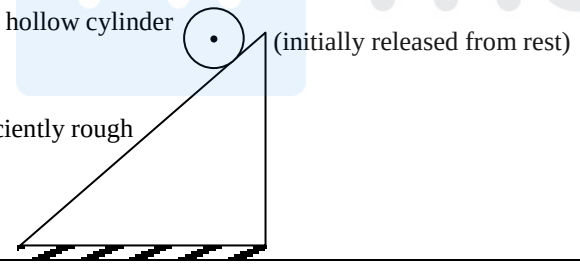
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1. The figure represent a sinusoidal wave whose general equation is $y = aA\sin(kx + b\omega t)$ where A, k, ω, are some arbitrary constants. P is a general point on the wave at a horizontal distance x from origin O, and x_1, x_2, x_3, x_4 are some special point on x-axis dividing wavelength in four equal intervals. V_p is the particle velocity of point P. Then match the following options ($|a| = |b| = 1$)



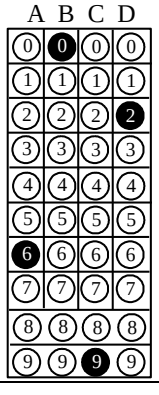
Column I		Column II	
(A)	V_P is +ve	**1.**	if $a > 0, b < 0, 0 < x < x_1$
(B)	V_P is –ve	**2.**	if $a < 0, b < 0, x_1 < x < x_2$
(C)	Wave is travelling along +ve x axis	**3.**	if $a > 0, b > 0, x_2 < x < x_3$
(D)	Wave is travelling along –ve x-axis	**4.**	if $a < 0, b > 0, x_2 < x < x_3$
		5.	if $a > 0, b < 0, x_1 < x < x_2$

2. Column – I shows few systems. In column – II few statements are given about those systems. Choose proper statement(s) for each system.

Column I	**Column II**
(A) Solid homogeneous disc 	**1.** friction acting along the direction of motion of centre of mass.
(B) Solid homogeneous sphere 	**2.** friction acting opposite to the direction of motion of centre of mass.
(C) 	**3.** friction acting is zero.
(D) 	**4.** Net work done by friction is zero.
	5. Net work done by friction is negative.

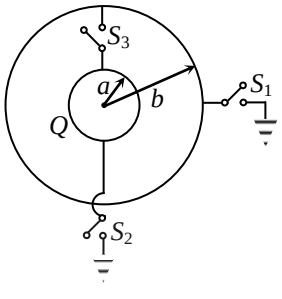
SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



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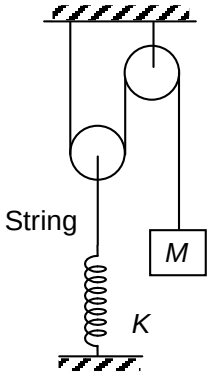
1. The figure shows a conducting sphere 'A' of radius 'a' which is surrounded by a neutral conducting spherical shell B of radius 'b' ($>a$). Initially switches S_1, S_2 and S_3 are open and sphere 'A' carries a charge Q. First the switch 'S_1' is closed to connect the shell B with the ground and then opened. Now the switch 'S_2' is closed so that the sphere 'A' is grounded and then S_2 is opened. Finally, the switch 'S_3' is closed to connect the spheres together. The heat (in Joule) which is produced after closing the switch S_3 is $0.2x$. Find x [Consider b = 4 cm, a = 2 cm and Q = 8 μC]



Space for work

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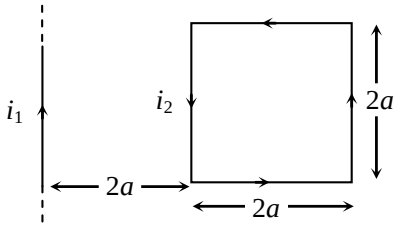
2. Consider the given system. All strings, pulleys and spring are ideal, mass M is in equilibrium. Maximum amplitude of oscillation of block in vertical direction is $\alpha\frac{Mg}{k}$. Find value of α.



Space for work

mathongo

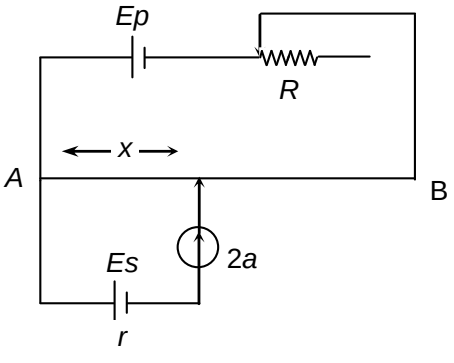
3. An infinitely long straight wire carries a current $i_1 = 1$ amp. A rectangular current loop placed in the same plane of shown dimensions carries a current $i_2 = 1$ amp anticlockwise. Magnitude of net force on the loop is found to be $\beta\frac{\mu_0}{4\pi}$. Find value of β.



Space for work

mathongo

4. Consider a potentiometer circuit. Primary cell is ideal. The length of potentiometer wire is 1m and the resistance per unit length of potentiometer wire varies with length as $\lambda = 2x\,\Omega/m$. Where x is distance from end A. Resistance of Rheostat varies with time as $R = t^2\Omega$. Null deflection point for secondary cell is obtained at $x = \frac{1}{2}$ m and at $t = 1$ sec. If emf of secondary cell is $\frac{1}{\gamma}$ times of emf of primary cell. Find γ.



Space for work

mathongo

CHEMISTRY

SECTION – I
Single Correct Choice Type

This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

---

1. A vessel filled with He gas at a pressure of 4 bar had a small thin orifice through which helium escaped into an evacuated space at the rate of 7.07 mmole/hr. How long would it take for 10 mmole of O_2 to leak through a similar orifice if O_2 were confined at same pressure and temperature? (Atomic mass of He = 4, O = 16)

(a) 2 hr (b) 3 hr
(c) 4 hr (d) 5 hr

2. When $NaNO_3$ (d = 2.0 g/cc) is heated in a closed vessel of 100 ml, oxygen is liberated and $NaNO_2$ (d = 1.5 g/cc) is left behind as per the reaction: $2NaNO_3(s) \rightleftharpoons 2NaNO_2(s) + O_2(g)$. At equilibrium, the volumes of $NaNO_3$ left and $NaNO_2$ produced are very small and can be neglected. Which of the following is a correct statement about this equilibrium?

(a) Addition of 30 g of $NaNO_2$ favours forward reaction.
(b) Addition of 30 g of $NaNO_3$ favours reverse reaction.
(c) Increasing temperature favours reverse reaction.
(d) None of these.

3. When 60 ml of 0.1M $Ca(NO_3)_2$ is mixed with 40 ml of 0.125M Na_2CO_3, $CaCO_3$ precipitates. If Ksp of $CaCO_3$ is 5×10^{-9} M^2, the $[CO_3^{2-}]$ in the resulting solution is

(a) 5×10^{-6} M (b) 5×10^{-7} M
(c) 5×10^{-8} M (d) 5×10^{-9} M

4. If 0.5 L of a 1 M $SnSO_4$ solution is electrolysed for a period of 30 minutes using a current of 9.65 A and inert electrodes, the final concentration of Sn^{2+} in the solution will be (Given: $E^{o}_{Sn^{2+}|Sn} = -0.14V$, $E^{o}_{Sn^{2+}|Sn^{4+}} = -0.13V$, $E^{o}_{OH^{-}|O_2} = -0.4V$, $E^{o}_{SO_4^{2-}|S_2O_8^{2-}} = -2.0V$)

(a) 0.64 M (b) 0.28 M

(c) 1.18 M (d) 0.84 M

5. The number of linkage isomers for the compound $K_2[Cu(CN)_2\ (NCS)_2]$ are (including it)

(a) 4 (b) 6

(c) 16 (d) 9

6. Which of the following is a correct statement about s-block compounds?

(a) Stability of metal chlorides decreases down the alkali metal group.

(b) Lattice energy of M_2O_2 is greater than that of M_2O.

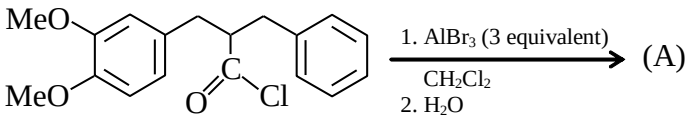
(c) Stability of peroxide increases with the increase in size of alkali metal cation.

(d) The water of crystallization is greater in alkali metal salts than in alkaline earth metal salts.

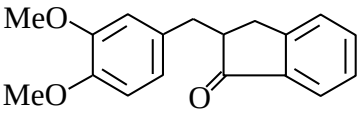
7. Amongst NO_3^-, AsO_3^{3-}, CO_3^{2-}, ClO_3^-, SO_3^{2-} and BO_3^{3-}, the non-planar species are (atomic no: N = 7, As = 33, C = 6, Cl = 17, S = 16, B = 5)

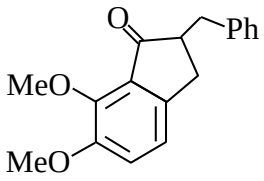
(a) CO_3^{2-}, SO_3^{2-} and BO_3^{3-} (b) AsO_3^{3-}, ClO_3^- and SO_3^{2-}

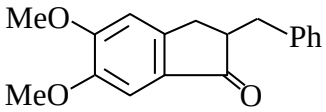
(c) NO_3^-, CO_3^{2-} and BO_3^{3-} (d) SO_3^{2-}, NO_3^- and BO_3^{3-}

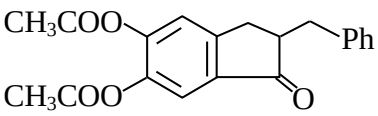
8. 

The product (A) would be

(a) 

(b) 

(c) 

(d) 

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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9. When radioactive sulfur is added to an alkaline sodium sulfite solution, radioactive thiosulphate ion is formed. Upon adding Ba^{2+}, a precipitate of BaS_2O_3 is formed. The precipitate is filtered, dried and then treated with acid, producing solid sulfur, SO_2 gas and water. The correct set of products is (* indicates radioactive sulfur)

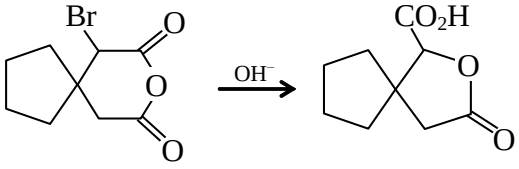
(a) $SO_2 + S_8$ (b) $SO_2 + \overset{*}{S}_8$

(c) $\overset{*}{S}O_2 + S_8$ (d) $\overset{*}{S}O_2 + \overset{*}{S}_8$

10. Which of the following oxyacid(s) of sulphur have S–S bond?

(a) $H_2S_2O_8$ (b) $H_2S_2O_4$

(c) $H_2S_2O_5$ (d) $H_2S_2O_6$

11. 

The given reaction involves:

(a) Nucleophilic substitution at acyl carbon.

(b) Nucleophilic substitution at alkyl carbon.

(c) Elimination

(d) Ring contraction

12. The amount of dissolved oxygen in 1 litre water in equilibrium with the air at 1 atm pressure at 25°C will be (assume that air contains 20 mole % oxygen, Henry's constant (k_H) for oxygen is 3.04×10^7 mm Hg and density of H_2O at 25°C is 1 g/cc)

(a) the mole fraction of O_2 in solution is 5×10^{-7}.

(b) the mole fraction of O_2 in solution is 5×10^{-6}.

(c) the molarity of O_2 in solution is 2.77×10^{-5} M

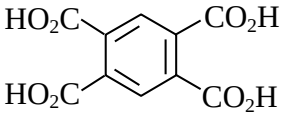
(d) the molarity of O_2 in solution is 2.77×10^{-4}M

SECTION–III
Comprehension Type

This section contains 1 group of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage–I

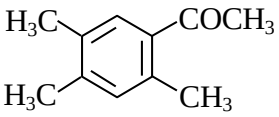
Compound (A), $C_{11}H_{12}O$, which gave no Tollen's test, was treated with $LiAlH_4$, followed by dilute acid, to give compound (B), which could be resolved into enantiomers. When optically active (B) was treated with CrO_3 in pyridine, an optically inactive sample of (A) was obtained. Heating (A) with hydrazine in base gave hydrocarbon (C), which, when heated with alkaline $KMnO_4$ followed by acidification, gave carboxylic acid (D).

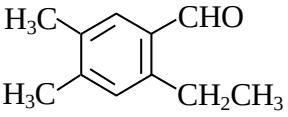


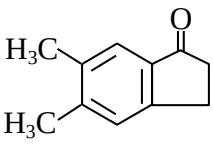
(D)

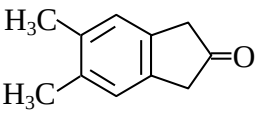
Another compound (X), $C_6H_{12}O_2$, was found to be optically active and was slowly oxidized to an optically active carboxylic acid (Y), $C_6H_{12}O_3$, by $[Ag(NH_3)_2]^+$. Oxidation of (X) by anhydrous CrO_3 gave an optically inactive compound that reacted with Zn amalgam/HCl to give 3-methylpentane. With aqueous H_2CrO_4, compound (X) was oxidized to an optically inactive dicarboxylic acid (Z), $C_6H_{10}O_4$.

13. The most probable structure of (A) will be

(a) 

(b) 

(c) 

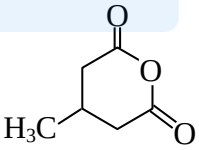
(d) 

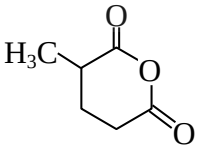
14. The structure of compound (X) would be

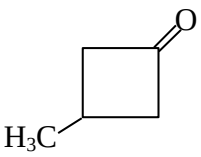
(a) $OHCCH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_2CHO$ (b) $OHCCH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–COCH_3$

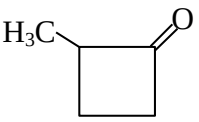
(c) $HOH_2CCH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–COCH_3$ (d) $OHCCH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_2CH_2OH$

15. Compound (Z) on heating with barium oxide forms

(a) 

(b) 

(c) 

(d) 

SECTION–IV

Matrix–Match Type

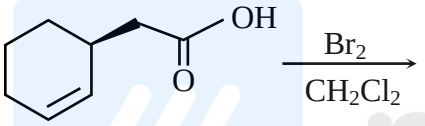
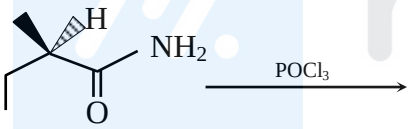
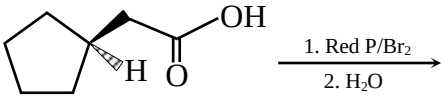
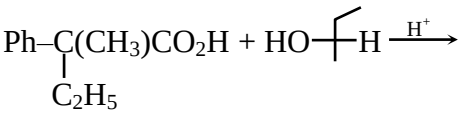
This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

	1	2	3	4	5
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

---

1. Match the following

Column I	**Column II**
(A) 	**1.** Product is a racemic mixture.
(B) 	**2.** Substrate can show optical isomerism.
(C) 	**3.** Product is optically active.
(D) 	**4.** Product is capable of showing geometrical isomerism. **5.** Diastereomers are formed.

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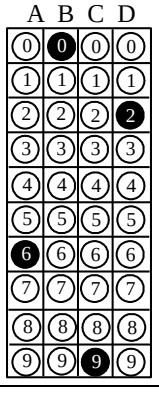
2. Match the following

Column I	**Column II**
(A) 2.0 g NaOH + 5.3 g Na_2CO_3 in 1 L H_2O	**1.** Vol. of 0.1M HCl used in presence of phenolphthalein indicator = 500 mL
(B) 1.0 g NaOH + 2.1 g $NaHCO_3$ in 1L H_2O	**2.** Vol. of 0.1 M HCl used in presence of methyl orange indicator = 500 mL
(C) 5.3 g Na_2CO_3 + 2.1 g $NaHCO_3$ in 1 L H_2O	**3.** Mass of NaCl produced in presence of phenolphthalein indicator = 5.85
(D) 1.06 g Na_2CO_3 in 1 L H_2O	**4.** Mass of NaCl produced in presence of methyl orange indicator = 1.17 g
	5. Mass of $CaCO_3$ precipitated after addition of excess of $CaCl_2$ solution to it. = 5 g.

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SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



---

1. A 40 ml sample of an aqueous solution of the methylamine at 25°C is titrated with 0.15 M HCl and the equivalence point is reached when 40 ml of the acid have been added. The pH at the equivalence point is ____ (Given : K_a for $CH_3NH_3^+$ is $\frac{4}{3}\times10^{-11}$).

Space for work

mathongo

2. The number of $R_2Si(OH)_2$ units required to prepare a silicone polymer containing 10 $Si-O-Si$ linkages is ——.

Space for work

mathongo

3. A compound with molecular formula $C_6H_{14}O_5$ is converted by the action of acetyl chloride to a compound with molecular mass 334. The number of OH groups in the compound is ——.

Space for work

mathongo

4. The ground state electronic wave function of hydrogen atom is proportional to e^{-r/a_0} where r is the distance from the nucleus and a_0 is a constant.

Calculate the relative probability of finding the electron inside a region of volume 1.0 pm^3, located at the nucleus to a distance of a_0 from the nucleus. Report your answer in single digit ignoring any fraction involved. (Given: ln 0.14 = – 2)

Space for work

mathongo

MATHEMATICS

SECTION – I
Single Correct Choice Type

This section contains 6 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

---

1. From a point P outside a circle with centre at C, tangents PA and PB are drawn such that $\frac{1}{(CA)^2}+\frac{1}{(PA)^2}=\frac{1}{16}$, then the length of chord AB is

(a) 8 (b) 12 (c) 16 (d) none of these

2. The coefficient of x^2 in $(1+x)(1+2x)(1+4x).....(1+2^n x)$ is

(a) $\frac{(2^n+1)(2^n+2)}{3}$ (b) $\frac{(2^{n+1}-1)(2^{n+1}-2)}{3}$

(c) $\frac{(2^{n+1}+1)(2^{n+1}+2)}{3}$ (d) $\frac{(2^{n+1}+1)(2^{n+1}-2)}{5}$

3. If the function $f(x)=\left(4\sin^2 x-1\right)^n\left(x^2-x+1\right), n\in N$ has a local minimum at $x=\frac{\pi}{6}$, then

(a) n can be any odd number
(b) n can only be an odd prime number
(c) n can be any even number
(d) n must be a multiple of four

4. Let $I_1=\int_0^1 \frac{e^x}{1+x}dx$ and $I_2=\int_0^1 \frac{x^2dx}{e^{x^3}(2-x^3)}$. Then $\frac{I_1}{I_2}$ is

(a) $\frac{3}{e}$ (b) $\frac{e}{3}$ (c) $3e$ (d) $\frac{1}{3e}$

5. Suppose $f(x) = x^3 + ax^2 + bx + c$, where *a*, *b*, *c* are chosen respectively by throwing a die three times. Then the probability that $f(x)$ is an increasing function, is

(a) $\frac{4}{9}$ (b) $\frac{3}{8}$ (c) $\frac{2}{5}$ (d) $\frac{16}{34}$

6. ΔABC is inscribed in a unit circle. The three bisectors of angles *A*, *B* and *C* are extended to intersect the circle at A_1, B_1 and C_1 respectively.

Then the value of $\frac{AA_1 \cos\frac{A}{2} + BB_1 \cos\frac{B}{2} + CC_1 \cos\frac{C}{2}}{\sin A + \sin B + \sin C}$ is

(a) 2 (b) 4

(c) 6 (d) 8

SECTION – II
Multiple Correct Choice Type

This section contains 5 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

---

7. The values of *a*, for which $\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$ doesn't have a real solution, are

(a) – 10 (b) 12 (c) 5 (d) – 30

8. If $f(x + y) = f(x) f(y)$ for all $x, y \in R$, $f(1) = 2$ and $a_r = f(r)$ for $r \in N$, then the co-ordinates of a point on the parabola $y^2 = 8x$ whose focal distance is 4, may be

(a) (a_1, a_2) (b) $(a_1, -a_2)$

(c) $(a_3, -a_3)$ (d) (a_3, a_3)

9. If $\int \frac{dx}{1-\sin^4 x} = \frac{1}{2}\tan x + A\tan^{-1}(f(x)) + C$, then

(a) $A = \sqrt{2}$ (b) $A = \frac{1}{2\sqrt{2}}$

(c) $f(x) = \frac{\tan x}{\sqrt{2}}$ (d) $f(x) = \sqrt{2}\tan x$

10. If the unit vectors $\vec{a}$ and $\vec{b}$ are inclined at an angle 2θ such that $|\vec{a} - \vec{b}| < 1$ and $0 \le \theta \le \pi$, then θ lies in the interval

(a) $\left[0, \frac{\pi}{6}\right)$ (b) $\left(\frac{5\pi}{6}, \pi\right]$

(c) $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$ (d) $\left[\frac{\pi}{2}, \frac{5\pi}{6}\right]$

11. Let $\alpha = \cos\frac{2\pi}{11} + i\sin\frac{2\pi}{11}$, $\lambda = \alpha^6$, $\mu = \alpha^7$, $\beta = \alpha^2$, then

(a) $\text{Re}(\lambda + \lambda^2 + \lambda^3 + \lambda^4 + \lambda^5) = -\frac{1}{2}$

(b) $(\mu - \beta)(\mu - \beta^2)(\mu - \beta^3)\ldots(\mu - \beta^9)(\mu - \beta^{10}) = 0$

(c) $(i - \beta)(i - \beta^2)(i - \beta^3)\ldots(i - \beta^{10}) = i$

(d) $(\alpha - \beta)(\alpha - \beta^2)\ldots\ldots(\alpha - \beta^9)(\alpha - \beta^{10}) = 1$

SECTION – III
Reasoning Type

This section contains 5 reasoning type questions numbered 12 to 16. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

12. **Statement-1** : $f(x)=[x]^2+\sqrt{\{x\}^2}$ is discontinuous at all integers (where [.] denotes greatest integer function and {.} denotes fractional part function).

Statement-2 : $f(x)=[x]$ is discontinuous at all integers.

(a) A (b) B (c) C (d) D

13. **Statement-1** : If positive numbers x, y, z satisfy $x + y + z = 1$, then $\left(\frac{(1-2x)+(1-2y)+(1-2z)}{3}\right) \geq ((1-2x)(1-2y)(1-2z))^{\frac{1}{3}}$

Statement-2 : For any 3 positive numbers a, b, c their A.M. ≥ G.M.

(a) A (b) B (c) C (d) D

14. **Statement-1** : Any positive integral power of $(\sqrt{2}-1)$ can be expressed as $\sqrt{N}-\sqrt{N-1}$ for some natural number $N > 1$.

Statement-2 : Any positive integral power of $(\sqrt{2}-1)$ can be expressed as $A+B\sqrt{2}$ where A and B are integers.

(a) A (b) B (c) C (d) D

15. **Statement-1** : Let $A = \{x \mid x$ is a prime number and $x < 30\}$. Then the number of different rational numbers, whose numerator and denominator belong to A is 93.

Statement-2 : $\frac{p}{q}$ is a rational number $\forall q \neq 0$ and $p, q \in I$.

(a) A (b) B (c) C (d) D

16. **Statement-1** : The number of solution(s) of the equation

$$(\sin x - 2\cos x) = \min\left\{\sqrt{3},\ a^2 - 4a + 5\right\} \text{ is 2.}$$

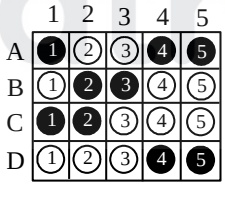
Statement-2 : Range of $\min\left(\sqrt{3},\ a^2 - 4a + 5\right)$ is $[1,\ \sqrt{3}]$ for $a \in R$.

(a) A (b) B (c) C (d) D

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1. $f(x)=\dfrac{2^x+2^{-x}}{2^x-2^{-x}}$

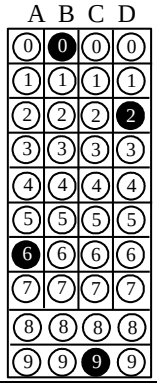
Column- I	Column -II
(A) If $1<x<2$, then $f(x)$ satisfies	**1.** $f(x)>1$
(B) If $0<x<\dfrac{1}{2}$, then $f(x)$ satisfies	**2.** $f(x)<-1$
(C) If $-\dfrac{1}{2}<x<0$, then $f(x)$ satisfies	**3.** $f(x)>3$
(D) If $-2<x<-1$, then $f(x)$ satisfies	**4.** $f(x)<-3$ **5.** $-1<f(x)<1$

2. A matrix of order 3 × 3 is formed by using the elements {1, 2, ……9} without repetition. Considering this; match column I and II (here $a = 2$, $b = 3$, $c = 5$)

Column- I	Column -II
(A) Probability that first row of the matrix has entries in increasing order, is	**1.** $\dfrac{1}{ab}$
(B) Probability that first row and first column have entries in increasing order, is	**2.** $\dfrac{1}{a^2c}$
(C) Probability that first row and first column have entries in increasing order given that last entry of the third row and third column is the greatest, is	**3.** $\dfrac{a^2+c}{a^2b^2c}$
(D) Probability that first row of matrix has entries in increasing order given that entries of second row and third row are in increasing order, is	**4.** $\dfrac{c-a}{ab^2}$ **5.** $\dfrac{1}{a}-\dfrac{1}{b}$

SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



1. Two lines $zi - \bar{z}\,i + 2 = 0$ and $z(1+i) + \bar{z}(1-i) + 2 = 0$ intersect at a point P. There is a complex number $\alpha = x + iy$ at a distance of 2 units from the point P which lies on line $z(1+i) + \bar{z}(1-i) + 2 = 0$. Find $[|x|]$ (where $[\cdot]$ represents greatest integer function).

Space for work

mathongo

2. Let P, Q be two points on the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ whose eccentric angles differ by a right angle. Tangents are drawn at P and Q to meet at R. If the chord PQ divides the join of C and R in the ratio $m : n$ (C being centre of ellipse), then find $m + n$ ($m : n$ is in simplified form).

Space for work

mathongo

3. Find the value of $\frac{29\int_0^1 \left(1-x^4\right)^7 dx}{4\int_0^1 \left(1-x^4\right)^6 dx}$.

Space for work

mathongo

4. If $f(x)=a\cos(\pi x)+b,\ f'\left(\frac{1}{2}\right)=\pi$ and $\int_{1/2}^{3/2} f(x)dx=\frac{2}{\pi}+1$, then find the value of $-\frac{12}{\pi}\left(\sin^{-1}a+\cos^{-1}b\right)$.

Space for work

mathongo

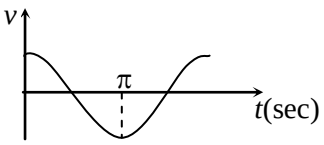
PAPER-II
PHYSICS
SECTION – I
Single Correct Choice Type

1. Length of the plate = 8.32 – (– 0.02) cm = 8.34 cm

$\therefore$ **(b)**

2. $x = \sin \omega t$

$v = \dfrac{dx}{dt} = \cos \omega t$



Since speed of particle decreases during

$(t = 0 \sec \text{ to } t = \dfrac{\pi}{2} \sec)$ and $(t = \pi \sec \text{ to } t = \dfrac{3\pi}{2} \sec)$

$\therefore$ **(b)**

3. When system will again come to rest the block will move $(l + 3l + l = 5l)$ distance on plank.

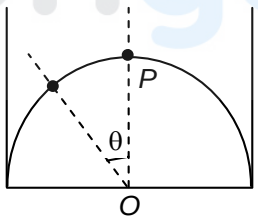
Let maximum displacement of plank be Δx

Since centre of mass remain at rest

$$\Delta x = \frac{5ml}{m + M}$$

$\therefore$ **(a)**

4. When the bead will reach at *P*, there will be maximum contact force between bead and structure. If the mass of bead is m and its speed at *P* is *v* then



Net work done by buoyancy and gravity = Δ K.E.

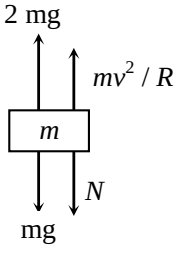
$$\text{mg } R(1 - \cos\theta) = \frac{1}{2}mv^2$$

$$mgR\left(2\sin^2\frac{\theta}{2}\right) = \frac{1}{2}mv^2$$

$$mgR\,2\frac{\theta^2}{4} = \frac{1}{2}mv^2$$

$$\Rightarrow \quad \frac{mv^2}{R} = mg\theta^2$$

Drawing *FBD* of bead at *P*.



$$N + mg = 2mg + \frac{mv^2}{R}$$

$$N = mg + \frac{mv^2}{R} = mg + mg\theta^2 = mg(1 + \theta^2)$$

$$N \,\alpha\, (1 + \theta^2)$$

$\therefore$ **(b)**

5. $\Delta Q = \Delta U + \Delta W$

$CdT = C_v dT + PdV$

$C = C_v + P\dfrac{dV}{dT}$ …(1)

Since $PV = RT$

$P\dfrac{dV}{dT} + V\dfrac{dp}{dT} = R$ …(2)

and $T^2 = 4P$

$2T = 4\dfrac{dP}{dT} \Rightarrow \dfrac{dP}{dT} = \dfrac{T}{2}$ …(3)

from (2) and (3) $P\dfrac{dV}{dT} + \dfrac{VT}{2} = R$

$P\dfrac{dV}{dT} + 2R = R$

$P\dfrac{dV}{dT} = -R$ …(4)

From (1) and (4) $C = C_v - R = \dfrac{3}{2}R - R = \dfrac{R}{2}$

$\therefore$ **(c)**

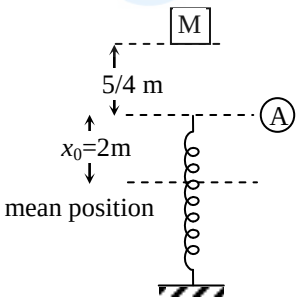
6. Say the block takes time t_1 to reach at A. $\dfrac{1}{2}gt_1^2 = \dfrac{5}{4} \Rightarrow t_1 = \dfrac{1}{2}$ sec

Now block will perform SHM.

Let at time of force balanced position compression in spring is x_0 then

$kx_0 = mg \Rightarrow 5x_0 = 10 \Rightarrow x_0 = 2\,m$

This will be the mean position of SHM. Maximum compression in the spring is if y_0 then

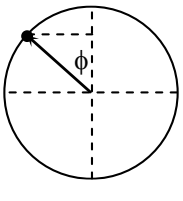


$mg\left(\dfrac{5}{4} + y_0\right) = \dfrac{1}{2}ky_0^2 \Rightarrow 10\left(\dfrac{5}{4} + y_0\right) = \dfrac{5}{2}y_0^2 \Rightarrow \quad y_0 = 5\text{m}$

Therefore block will perform SHM with amplitude $A = (5-2)\,\text{m} = 3\text{m}.$ with angular frequency $\omega = \sqrt{\dfrac{k}{m}} = \sqrt{5}$ mean position is shown. Let block performs SHM for time T.

Drawing phasor of SHM.

$3\cos\phi = 2 \Rightarrow \cos\phi = \dfrac{2}{3} \Rightarrow \phi = \cos^{-1}\left(\dfrac{2}{3}\right)$



So $T = \dfrac{2\pi - 2\phi}{\omega} = \dfrac{2\pi}{\sqrt{5}} - \dfrac{2}{\sqrt{5}}\cos^{-1}\left(\dfrac{2}{3}\right)$

So overall time it takes to gain its initial position again $= 2t_1 + T$

$= 1 + \frac{2\pi}{\sqrt{5}} - \frac{2}{\sqrt{5}}\cos^{-1}\left(\frac{2}{3}\right)$

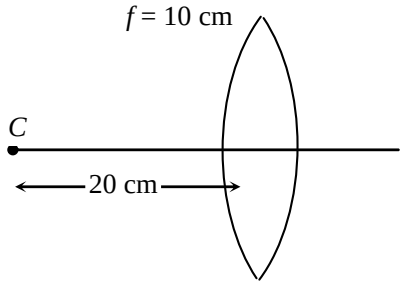
$\therefore$ **(b)**

7. **Case – 1**

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} - \frac{1}{(-20)} = \frac{1}{10} \Rightarrow \frac{1}{v} = \frac{2}{20} - \frac{1}{20} \Rightarrow v = 20\,\text{cm}$

Magnification $= \frac{v}{u} = \frac{20}{20} = -1 = m_1$

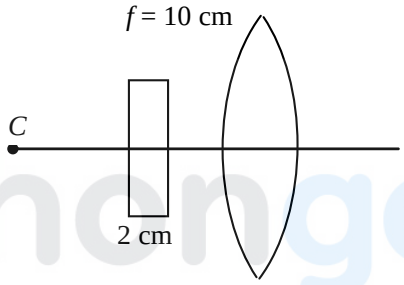


Case – 2

$u = -\left(18 + \frac{2}{\mu}\right)$

$\frac{1}{v} - \frac{1}{-\left(18 + \frac{2}{\mu}\right)} = \frac{1}{10}$

$\Rightarrow \frac{1}{v} = \frac{1}{10} - \frac{1}{\left(18 + \frac{2}{\mu}\right)}$



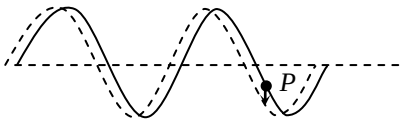
$\Rightarrow \frac{1}{v} = \frac{18 + \frac{2}{\mu} - 10}{10\left(18 + \frac{2}{\mu}\right)} = \frac{8 + \frac{2}{\mu}}{10\left(18 + \frac{2}{\mu}\right)}\,\text{cm.}$

Magnification $m_2 = \frac{\frac{10\left(18 + \frac{2}{\mu}\right)}{\left(8 + \frac{2}{\mu}\right)}}{-\left(18 + \frac{2}{\mu}\right)} = -\frac{1}{2} = \frac{m_1}{2}$

$\Rightarrow \frac{10}{8 + \frac{2}{\mu}} = \frac{1}{2} \Rightarrow 20 = 8 + \frac{2}{\mu} \Rightarrow \frac{2}{\mu} = 12 \Rightarrow \mu = \frac{1}{6} = 0.166 = 0.17$

$\therefore$ **(b)**

8. Drawing the shape of string after sometime.



Its clear that velocity of P at this instant is downwards.

$\therefore$ **(b)**

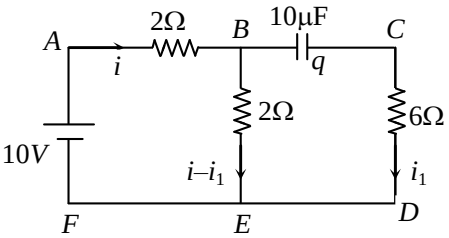
SECTION – II
Multiple Correct Choice Type

9. for loop $ABEFA$

$10 = 2i + 2(i - i_1) \Rightarrow 5 = 2i - i_1$ …(1)

for loop $BCDEB$

$2(i - i_1) = \frac{q}{C} + 6i_1 \Rightarrow 2i = \frac{q}{C} + 8i_1$ …(2)



from (1) and (2)

$5 + i_1 = \frac{q}{C} + 8i_1 \Rightarrow 7i_1 + \frac{q}{C} = 5$ since $i_i = \frac{dq}{dt}$

or $7\frac{dq}{dt} + \frac{q}{C} = 5$

$7\frac{dq}{dt} = \frac{5C - q}{C}$

$\int_0^q \frac{dq}{5C - q} = \int_0^q \frac{dt}{7C} \Rightarrow -\left[\ln(5C - q)\right]_0^q = \frac{t}{7C}$

$\Rightarrow \frac{5C - q}{5C} = e^{-\frac{t}{7C}}$

$q = 5C\left(1 - e^{-t/7C}\right) = q_0\left(1 - e^{-t/7C}\right)$

Time constant $\tau = 7C = 7 \times 10 \times 10^{-6}$ sec $= 70 \times 10^{-6}$ sec. $= 70\mu$sec.

In steady state charge on capacitor $q = q_0 = 5C = 50\mu C$ at any instant current through 6Ω is $i_1 = \frac{5}{7}e^{-t/\tau}, \tau = 70\mu$sec.

$\therefore$ **(a, b, c)**

10. $N = mg + qE$ but $\frac{q}{m} = 8 \Rightarrow q = 8m$

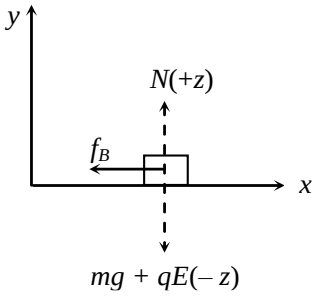
$N = mg + 8mE$

$= 50m$

$f = \mu N = \frac{50}{3}m$

$a_t = \frac{f}{m} = -\frac{50}{3}$ m/s

Since $v = u + a_t t$



$$0 = 100 - \frac{50}{3}t \Rightarrow t = 6\,\text{sec}$$

$$s = ut + \frac{1}{2}a_t t^2$$

$$s = 100(6) - \frac{1}{2}\left(\frac{50}{3}\right)(6)^2 = s = 300\ m$$

Since force due to electric and magnetic field on the particle is perpendicular to the displacement so work done by electric field and magnetic field is zero.

∴ **(a, b, c, d)**

11. For the considered region given charge distribution will act as a dipole system. Therefore electric field anywhere in the region is directed towards +ve *x*-axis.

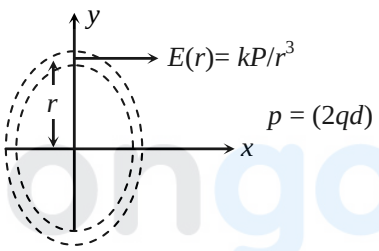
Due to symmetry we can conclude that electric potential anywhere in *y*-*z* plane will be zero.

So the work done from bringing a positive test change from $\left(0, \frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ to $\left(a, \frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ is zero.

Electric flux crossing

$$\phi = \int_{r=a}^{b} E(r)(2\pi r dr) = (kP)(2\pi)\int_a^b \frac{dr}{r^2}$$

$$= \frac{1}{4\pi \in_0}(2qd)(2\pi)\left(\frac{-1}{r}\right)_a^b = \frac{qd}{\in_0}\left(\frac{1}{a} - \frac{1}{b}\right)$$



∴ **(a, b, c, d)**

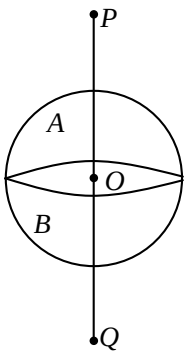
12. Lets assume the complete sphere. Upper hemisphere is denoted by *A* and lower by *B*.

Combined field intensity at *P* or *Q* is

$$E = \frac{GM}{(2R)^2} = \frac{G}{4R^2}\frac{4}{3}\pi R^3 \rho = \frac{\pi G \rho R}{3}$$

Its obvious that intensity at *P* and *Q* will be unlike parallel.

Its clear that if magnitude of intensity at *Q* due to only the hemisphere *B* is E_0 then at *P* it will be $\frac{\pi}{3}G\rho R - E_0$



∴ **(b, d)**

SECTION–III
Comprehension Type

13. $\frac{1}{3}$ mole of O_2 dissociates into atoms.

$\frac{2}{3}$ mole of O formed due to dissociation.

∴ No. of moles of monoatomic gas in mixture = $1 + \frac{2}{3} = \frac{5}{3}$ mole

$\therefore$ No. of moles of diatomic gas in mixture = $\frac{2}{3}$ mole

$$\frac{\mu_{mix}}{\gamma_{mix-1}} = \frac{\frac{5}{3}}{\frac{5}{3}-1} + \frac{\frac{2}{3}}{\frac{7}{5}-1}$$

$\gamma_{mix} = 1.56$

$\therefore$ **(a)**

14. For backward motion of piston,

$$TV^{\gamma-1} = T_0V_0^{\gamma-1}$$

$$T = T_0\left(\frac{V_0}{V_0/2}\right)^{1.56-1} = \sqrt{2}\,T_0$$

$\therefore$ **(a)**

15. $\Delta U = \mu_{mix}\frac{R}{\gamma_{mix-1}}(T-T_0) = \frac{25}{6}R\left(\sqrt{2}-1\right)T_0 = 14.18\ T_0$

$\therefore$ **(a)**

SECTION–IV
Matrix–Match Type

1. **A – (4, 5); B – (1, 2, 3); (C) – 1, 2, 5; (D) – (3, 4)**

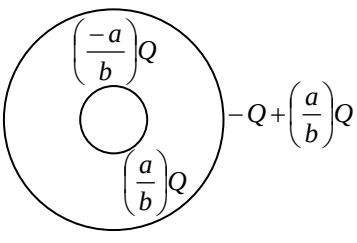
2. **A – (1, 4); B – (3, 4); C – (1, 5); D – (2, 4)**

SECTION–V
Integer Answer Type

1. When outer surface is grounded charge '$-Q$' resides on the inner surface of sphere 'B'

Now sphere A is connected to earth potential on its surface becomes zero.

Let the charge on the surface A becomes q



$$\frac{kq}{a} - \frac{kQ}{b} = 0$$

$$\Rightarrow \quad q = \frac{a}{b}Q$$

Consider the figure. In this position energy stored

$$E_1 = \frac{1}{8\pi\varepsilon_0 a}\left[\frac{a}{b}Q\right]^2 + \frac{Q^2}{8\pi\varepsilon_0 b} + \frac{1}{4\pi\varepsilon_0 b}\left[\frac{a}{b}Q\right](-Q)$$

when 'S_3' is closed, total charge will appear on the outer surface of shell 'B'. In this position energy stored $E_2 = \frac{1}{8\pi\varepsilon_0 b}\left(\frac{a}{b}-1\right)^2 Q^2$

Heat produced = $E_1 - E_2 = \dfrac{Q^2 a(b-a)}{8\pi\varepsilon_0 b^3} = 1.8$

$\therefore$ **9.**

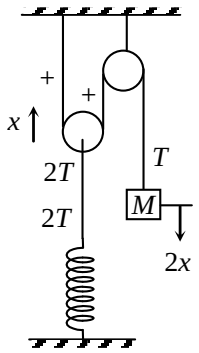
Ans. 9

2. Let elongation in the spring is x_0 when M is in equilibrium

$T = Mg$

and $kx_0 = 2T = 2Mg$

$x_0 = \dfrac{2Mg}{k}$



So maximum amplitude = $2x_0 = \dfrac{4Mg}{k}$

$\therefore \quad \alpha = 4$

Ans. 4

3. $F_{net} = F_1 - F_2$

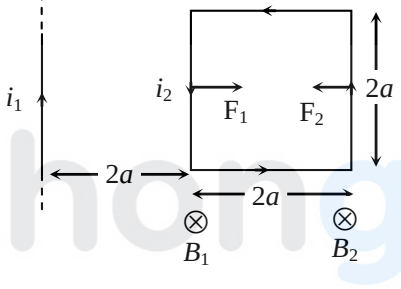
$= i_2 B_1(2a) - i_2(B_2)(2a)$

$= \left(\dfrac{\mu_0}{4\pi}\dfrac{2i_1}{2a}\right) i_2(2a) - \left(\dfrac{\mu_0}{4x}\dfrac{2i_1}{4a}\right) i_2(2a)$

$= \dfrac{2\mu_0}{4\pi} - \dfrac{\mu_0}{4\pi} = \dfrac{\mu_0}{4\pi}$

$\therefore \quad \beta = 1$

Ans. 1



4. Resistance of potentiometer wire $R' = \int_0^1 \lambda dx = 2\int_0^1 x dx = 1\Omega$

$\Rightarrow \quad E_S = \left(\dfrac{E_p}{R'+R}\right).\int_0^{\frac{1}{2}} \lambda dx = \left(\dfrac{E_p}{1+t^2}\right)\dfrac{1}{4}$

$E_S = \left(\dfrac{E_P}{1+1}\right)\dfrac{1}{4} = \dfrac{E_P}{8}$

$\therefore \quad \gamma = 8$

Ans. 8

CHEMISTRY

SECTION – I

Straight Objective Type

1. (c)

$$\frac{r_{He}}{r_{O_2}} = \frac{p_{He}}{p_{O_2}}\sqrt{\frac{M_{O_2}}{M_{He}}}$$

$$\frac{n_{He}/t_{He}}{n_{O_2}/t_{O_2}} = \frac{p_{He}}{p_{O_2}}\sqrt{\frac{M_{O_2}}{M_{He}}}$$

$$\frac{7.07 \times t_{O_2}}{1 \times 10} = \sqrt{\frac{32}{4}} = \sqrt{8} = 2\sqrt{2}$$

$$t_{O_2} = \frac{2 \times 14.14}{7.07} = \textbf{4 hr}$$

2. (b) Let the moles of $NaNO_3$, $NaNO_2$ and O_2 at equilibrium be a, b and c respectively and v_1 and v_2 will be the volume occupied by 'a' moles of $NaNO_3$ and 'b' moles of $NaNO_2$ respectively. They are small and can be ignored with respect to total volume i.e. (100 ml).

$$2NaNO_3(s) \rightleftharpoons 2NaNO_2(s) + O_2(g)$$

Moles at equilib.	a	b	c
Conc. at equilib.	$\frac{1000\,a}{v_1}$	$\frac{1000\,b}{v_2}$	$\frac{1000\,c}{100} = 10\,c = K_c$
Conc. after addn. of 30 g of $NaNO_2$ (v = 20 cc)	same	same	$\frac{1000\,c}{80} = \frac{100\,c}{8} = Q_c$, $Q_c > K_c$
Conc. after addn. of 30 g of $NaNO_3$ (v = 15 cc)	same	same	$\frac{1000\,c}{85} = Q_c'$, $Q_c' > K_c$

In both cases (addition of $NaNO_2$ as well as $NaNO_3$) the reaction will go in backward direction.

3. (b)

$$Ca(NO_3)_2(aq) + Na_2CO_3(aq) \longrightarrow CaCO_3(s) + 2NaNO_3(aq)$$

Initial millimoles	6	5	–	–
Millimoles after the reaction	1	0	5	10

$$CaCO_3(s) \rightleftharpoons Ca^{2+} + CO_3^{2-}$$

$$\frac{x+1}{100} \quad \frac{x}{100}$$

$$\approx \frac{1}{100}$$

$$Ksp = \left[Ca^{2+}\right]\left[CO_3^{2-}\right]$$

$5\times10^{-9} = \frac{1}{100}\times\left[CO_3^{2-}\right]$

$\left[CO_3^{2-}\right] = \mathbf{5\times10^{-7}M}$

4. **(a)** The reactions occuring during the electrolysis are

Cathode: $Sn^{2+} + 2e^- \longrightarrow Sn$

Anode: $Sn^{2+} \longrightarrow Sn^{4+} + 2e^-$

$2H^+ + 2e^- \longrightarrow H_2$

$$E_{H^+|H_2} = E^o_{H^+|H_2} - \frac{0.059}{2}\log\frac{(p_{H_2})}{[H^+]^2} = 0 - \frac{0.059}{2}\log\frac{1}{(10^{-7})^2}$$

$$= -0.059 \log (10^7) = -0.059 \times 7 = -0.413 \text{ V}$$

Moles of e^- used at cathode = Moles of e^- released at anode = $\frac{9.65\times30\times60}{96500} = 0.18$

Moles of Sn^{2+} oxidised = Moles of Sn^{2+} reduced = $\frac{0.18}{2} = 0.09$

Moles of Sn^{2+} left = 0.5 – 0.18 = 0.32

Concentration of Sn^{2+} left in solution = $\frac{0.32}{0.5}$ = **0.64 M**.

5. **(d)** The isomers are $[Cu\,(CN)_2\,(NCS)_2]^{2-}$

$[Cu\,(CN)\,(NC)\,(NCS)_2]^{2-}$

$[Cu\,(NC)_2\,(NCS)_2]^{2-}$

$[Cu\,(CN)_2\,(SCN)\,(NCS)]^{2-}$

$[Cu\,(CN)\,(NC)\,(SCN)\,(NCS)]^{2-}$

$[Cu\,(NC)_2\,(SCN)\,(NCS)]^{2-}$

$[Cu\,(CN)_2\,(SCN)_2]^{2-}$

$[Cu\,(CN)\,(NC)\,(SCN)_2]^{2-}$

$[Cu\,)(NC)_2\,(SCN)_2]^{2-}$

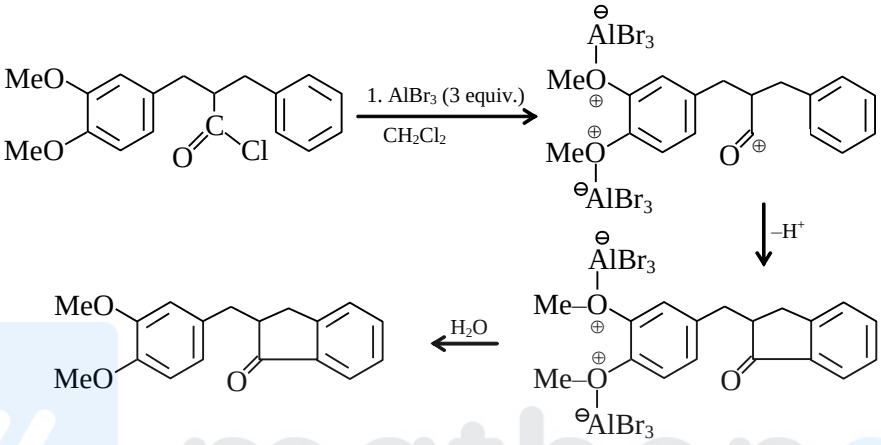
There are a total of **9** linkage isomers.

6. **(c)** O^{2-} is smaller than O_2^{2-}, thus, with a given metal ion (M^+), the lattice energy of M_2O would be greater than that M_2O_2. Peroxides would be preferably formed with larger cations because smaller cations will form a stable lattice of M_2O. Stability of metal chlorides, bromides and iodides increases down the group while that for fluorides decreases. The size of alkaline earth metal ions is smaller than that of alkali metal ions, so water of crystallization units, of alkaline earth metal salts are higher than that of alkali metal salts.

7. (b)

	NO_3^-	AsO_3^{3-}	CO_3^{2-}	ClO_3^-	SO_3^{2-}	BO_3^{3-}
$\frac{N}{2}$ value	3	4	3	4	4	3
No. of lone pair(s)	0	1	0	1	1	0
Shape	Trigonal planar	Trigonal pyramidal	Trigonal planar	Trigonal pyramidal	Trigonal pyramidal	Trigonal planar

8. (a)



One equivalent of $AlBr_3$ will be used to generate acylium ion and rest 2 equivalents react with MeO group to deactivate the ring (by forming Lewis acid-base complex having +ve charge on oxygen atom). Thus, acylium ion formed attacks on the other benzene ring (without MeO groups).

SECTION – II
Reasoning Type

9. (b)

$$SO_3^{2-} + \frac{1}{8}\overset{*}{S}_8 \longrightarrow \overset{*}{S}SO_3^{2-} \quad ; \quad {}^{-}O\!-\!\underset{\underset{\overset{\|}{S^*}}{\|}}{\overset{\overset{O}{\|}}{S}}\!-\!O^{-}$$

$$Ba^{2+} + \overset{*}{S}SO_3^{2-} \longrightarrow Ba\overset{*}{S}SO_3$$

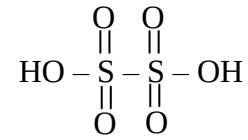
$$Ba\overset{*}{S}SO_3 + 2H_3O^+ \longrightarrow SO_2 + \frac{1}{8}\overset{*}{S}_8 + 3H_2O + Ba^{2+}$$

This reaction sequence indicate that the two sulphur atoms in thiosulphate are not equivalent and do not exchange with each other.

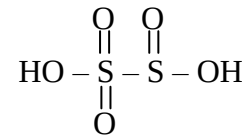
10. (b, c, d)

$H_2S_2O_4$: $HO - \underset{\|}{\overset{O}{S}} - \underset{\|}{\overset{O}{S}} - OH$

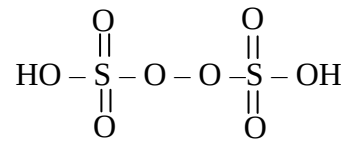
$H_2S_2O_6$:



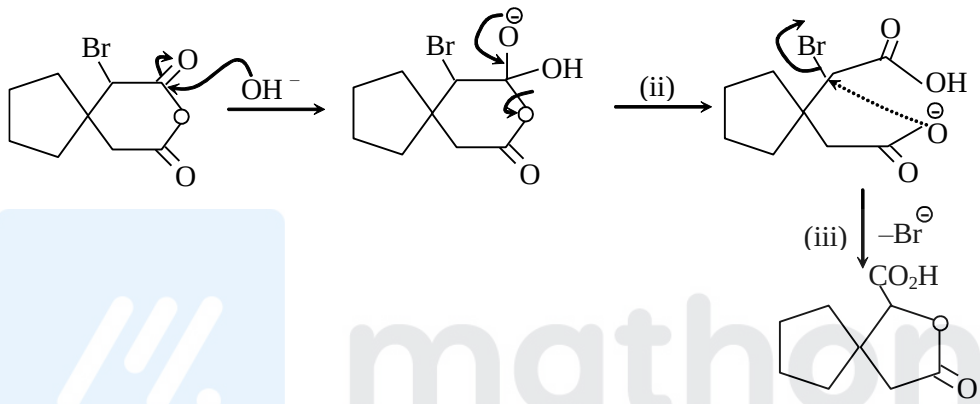
$H_2S_2O_5$:



$H_2S_2O_8$:



11. (a, b, d)



The first step of the reaction leads to nucleophilic substitution at acyl carbon while third step of the reaction involve nucleophilic substitution at alkyl carbon. Overall reaction leads to ring contraction from 6 to a 5-membered ring.

12. According to Dalton's law,

$p_{O_2} = P_{total} X_{O_2}^{gas} = 0.20 \times 760 = 152 \text{ mm Hg}$.

According to Henry's law,

$$p_{O_2} = k_H X_{O_2}^{liq}$$

$$152 = 3.04 \times 10^7 X_{O_2}^{liq}$$

$$X_{O_2} = 5 \times 10^{-6}$$

$$X_{O_2} = \frac{n_{O_2}}{n_{O_2} + n_{H_2O}} = \frac{n_{O_2}}{n_{O_2} + 1000/18} = \frac{n_{O_2}}{n_{O_2} + 55.56} \approx \frac{n_{O_2}}{55.56}$$

$\therefore$ Number of moles of O_2 dissolved $(n_{O_2}) = 55.56 \times 5 \times 10^{-6} = 2.77 \times 10^{-4}$

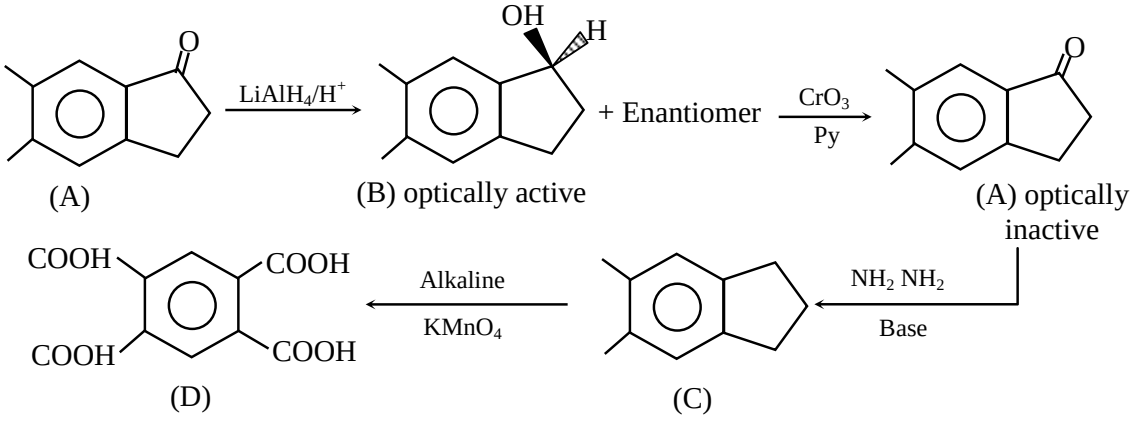
$[O_2] = 2.77 \times 10^{-4}$ M

$\therefore$ **(b) and (d)**

SECTION–III
Comprehension Type

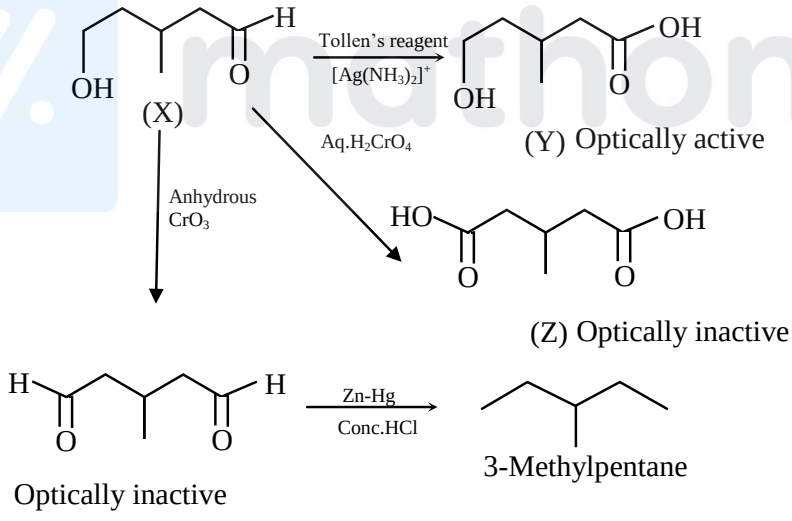
13. **(c)** Double bond equivalents of $C_{11}H_{12}O = 6$

No Tollen's test : Aldehyde absent, ketone may be present.

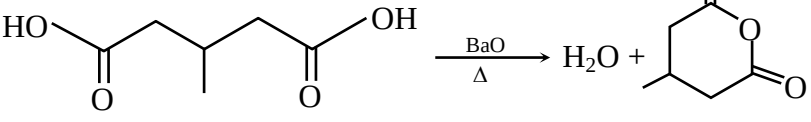


14. **(d)** $C_6H_{12}O_2(X)$ double bond equivalent= 1; Give positive Tollen's test.

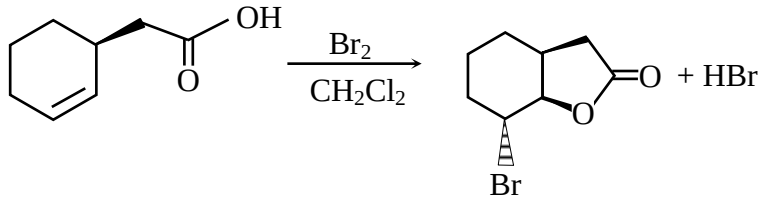
$\therefore$ Probable structure is

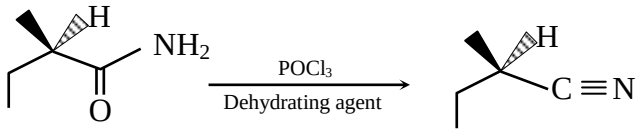


15. **(a)**

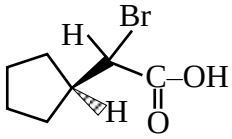


SECTION–IV
Matrix–Match Type

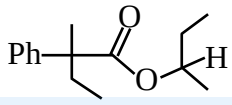
1. A - 

B - 

Optically active

C - 

racemic mixture

D - 

Diastereomeric mixture

Ans. (A): 2, 3, 4; (B): 3, 4; (C): 1; (D): 2, 3, 5

2. When phenolphthalein is used as indicator :

(A) Equivalent of HCl = Equivalent of NaOH + Equivalent of Na_2CO_3

$$= \frac{2}{40} + \frac{5.3}{106} = \frac{2}{20}$$

$\therefore$ Mass of NaCl $= \frac{2}{20} \times 58.5 = 5.85$

$\therefore$ No. of equivalent of HCl = $N_1\ V_1 \times 10^{-3} = \frac{2}{20}$

$\therefore$ $V_{HCl} = \frac{2}{200 \times 0.1 \times 10^{-3}} = \frac{2 \times 10 \times 1000}{200} = 100\,mL$

(B) No. of equivalent of HCl = $N_1\ V_1 \times 10^{-3} = \frac{1}{40}$

$\therefore$ $V_{HCl} = \frac{1}{40} \times \frac{1}{0.1} \times \frac{1}{10^{-3}} = \frac{10 \times 1000}{40} = 250\ mL$

(C) No. of equivalent of HCl = $\frac{5.3}{106} = \frac{1}{20}$

$\therefore$ $V_{HCl} = \frac{1}{20} \times \frac{1}{0.1} \times \frac{1}{10^{-3}} = \frac{10 \times 1000}{20} = 500\,mL$

(D) No. of equivalent of HCl = $\frac{1.06}{106} = \frac{1}{100}$

$\therefore$ $V_{HCl} = \frac{1}{100} \times 10 \times 1000 = 100\,mL$

When methyl orange is used as indicator:

(A) No. of equivalent of HCl = $= \frac{2}{40} + \frac{5.3}{53} = \frac{1}{20} + \frac{1}{10} = \frac{3}{20}$

$\therefore \quad V_{HCl} = \frac{3}{20} \times 10 \times 1000 = 1500 \text{ mL}$

(B) No. of equivalent of HCl = $\frac{1}{40} + \frac{2.1}{84} = \frac{2}{40} = \frac{1}{20}$

$\therefore \quad V_{HCl} = \frac{1}{20} \times 10 \times 1000 = 500 \text{ mL}$

(C) No. of equivalent of HCl = $\frac{5.3}{106} + \frac{2.1}{84} = \frac{5}{40}$

$\therefore \quad V_{HCl} = \frac{5}{40} \times 10 \times 1000 = 1250 \text{ mL}$

(D) No. of equivalent of HCl = $\frac{1.06}{53} = \frac{1}{50}$;

$\therefore$ Mass of NaCl = $\frac{1}{50} \times 58.5 = 1.17 \text{ g}$

$\therefore \quad V_{HCl} = \frac{1}{50} \times 10 \times 1000 = 200 \text{ mL}$

Mass of $CaCO_3$ formed :

$Na_2CO_3 + CaCl_2 \rightarrow CaCO_3 + 2NaCl$

$\therefore$ Moles of Na_2CO_3 = Moles of $CaCO_3$

$\therefore$ In (A), moles of $CaCO_3$ formed = $\frac{5.3}{106} = \frac{1}{20}$

$\therefore \quad W_{CaCO_3} = \frac{100}{20} = 5 \text{ g}$

In (C), Mole of $CaCO_3$ formed = $\frac{100}{20}$

$\therefore \quad W_{CaCO_3} = \frac{100}{20} = 5\text{g}$

In (D), $W_{CaCO_3} = \frac{1.06}{106} \times 100 = 1\text{g}$

Ans. **(A): 3, 5; (B) : 2; (C) : 1, 5 ; (D) : 4**

SECTION–V
Integer Answer Type

1.

$$CH_3NH_2 + HCl \longrightarrow CH_3\overset{\oplus}{N}H_3\,Cl^-$$

At equivalence point – – 40×0.15 = 6 mmole

$CH_3\overset{\oplus}{N}H_3$ undergo hydrolysis while Cl^- remain unhydrolysed.

$$CH_3NH_3^+ + H_2O \rightleftharpoons CH_3NH_2 + H_3O^+$$

$$\frac{6}{80}(1-h) \qquad \frac{6}{80}h \qquad \frac{6}{80}h$$

$$K_a = \frac{[CH_3NH_2]\,[H_3O^+]}{[CH_3NH_3^+]} = \frac{\frac{6}{80}h^2}{(1-h)} \simeq \frac{6}{80}h^2$$

$$h = \sqrt{\frac{K_a \times 80}{6}}$$

$$[H_3O^+] = \frac{6}{80}h = \sqrt{K_a \times \frac{6}{80}} = \sqrt{\frac{4}{3}\times 10^{-11} \times \frac{6}{80}} = \sqrt{10^{-12}}$$

$$[H_3O^+] = 10^{-6}\,M$$

$$\therefore\ pH = 6.0$$

2. To prepare a silicone containing 'n' Si–O–Si linkages, the total number of units required are (n+1), of which 2 are chain blocking units, R_3SiOH and (n–1) are chain building units, $R_2Si(OH)_2$. Thus, the number of $R_2Si(OH)_2$ units needed are **9.**

3. $R–OH + CH_3COCl \longrightarrow R–OCOCH_3 + HCl$

When an alcohol containing one OH group reacts with 1 mole of CH_3COCl, the molecular mass of the product will be 42 units more than that of ROH. If the original compound contains 'n' OH groups, then the molar mass of product will be 42n units more than that of original compound. In the present case,

166 + 42n = 334

$\therefore$ **n = 4**

Thus, the compound contains 4 – OH groups.

4. $\because$ dv = 1.0 pm^3

$\therefore\ P \propto \psi^2\,dv$, where $\psi^2 \propto e^{-2r/a_0}$

At nucleus, r = 0.

$\therefore\ P_{r=0} \propto e^0 . (1\ pm)^3 = 1\ (pm)^3$

At a distance $r = a_0$,

$$P_{r=a_0} \propto e^{-2}.(1.0\ pm^3) = 0.14 \times 1 pm^3$$

$$\therefore\ \frac{P_{r=0}}{P_{r=a_0}} = \frac{1}{0.14} = 7.1 = \mathbf{7}$$

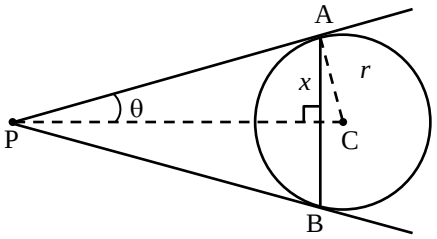
MATHEMATICS

SECTION – I

Single Correct Choice Type

1. **(a)** $\tan\theta = \dfrac{r}{PA}$

Given $\dfrac{1}{r^2} + \dfrac{1}{PA^2} = \dfrac{1}{16}$



$\Rightarrow \quad \dfrac{\cot^2\theta + 1}{(PA)^2} = \dfrac{1}{16}$

$\Rightarrow \quad PA\sin\theta = 4 = x \quad \Rightarrow \quad 2x = 8$

2. **(b)** Coefficient of $x^2 = \dfrac{1}{2}\left[\left(1 + 2 + 2^2 ..... + 2^n\right)^2\right] - \left[1^2 + 2^2 + ... \left(2^n\right)^2\right]$

$$= \frac{1}{2}\left[\left(\frac{2^{n+1}-1}{1}\right)^2 - \left(\frac{\left(2^2\right)^{n+1}-1}{3}\right)\right]$$

$$= \frac{1}{2}\left[\left(2^{n+1}-1\right)\left[\left(2^{n+1}-1\right) - \frac{2^{n+1}+1}{3}\right]\right]$$

$$= \frac{\left(2^{n+1}-1\right)\left(2^{n+1}-2\right)}{3}$$

3. **(c)** $f(x) = \left(4\sin^2 x - 1\right)^n \left(x^2 - x + 1\right)$

$f'(x) = n\left(4\sin^2 x - 1\right)^{n-1}(8\sin x\cos x)\left(x^2 - x + 1\right) + \left(4\sin^2 x - 1\right)^n (2x - 1)$

$= \left(4\sin^2 x - 1\right)^{n-1}[n\left(x^2 - x + 1\right).4\sin 2x + (2x - 1)\left(4\sin^2 x - 1\right)]$

If f has local minimum at $x = \dfrac{\pi}{6}$, then

$f'\left(\dfrac{\pi}{6}\right) = 0$, $f'\left(\dfrac{\pi}{6}^-\right) < 0$ and $f'\left(\dfrac{\pi}{6}^+\right) > 0$,

which is possible if $(n-1)$ is odd i.e. n even number.

4. **(c)** $I_2 = \dfrac{1}{3}\displaystyle\int_0^1 \frac{3x^2 dx}{e^{x^3}(2 - x^3)}$ put $t = x^3$

$I_2 = \dfrac{1}{3}\displaystyle\int_0^1 \frac{dt}{e^t(2-t)}$; $I_2 = \dfrac{1}{3}\displaystyle\int_0^1 \frac{dt}{e^{1-t}(1+t)}$

$\Rightarrow \quad \dfrac{I_1}{I_2} = 3e$

5. **(a)** $f'(x) = 3x^2 + 2ax + b$

$f(x)$ is increasing

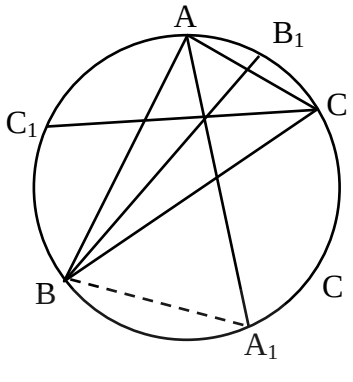
$\therefore$ $f'(x) \geq 0 \ \forall x$ where equality holds for discrete points.

$\therefore$ $D = (2a)^2 - 4 \times 3 \times b \leq 0 \ \Rightarrow \ a^2 - 3b \leq 0$

This is true for exactly 16 ordered pairs

$\therefore$ $\frac{16}{36} = \frac{4}{9}$.

6. **(a)** Join BA_1 as shown in diagram then



By sine rule for ΔABA_1

$AA_1 = 2\sin\left(B + \frac{A}{2}\right) = 2\cos\left(\frac{B}{2} - \frac{C}{2}\right)$

Thus $AA_1 \cos\frac{A}{2} = 2\cos\left(\frac{B-C}{2}\right)\cos\frac{A}{2} = \sin C + \sin B$

Similarly $BB_1 \cos\frac{B}{2} = \sin A + \sin C$

and $CC_1 \cos\frac{C}{2} = \sin A + \sin B$

So, $\frac{AA_1 \cos\frac{A}{2} + BB_1 \cos\frac{B}{2} + CC_1 \cos\frac{C}{2}}{\sin A + \sin B + \sin C} = 2$

SECTION – II
Multiple Correct Choice Type

7. **(b), (c), (d)**

Let $f(x) + \frac{a}{30} = 0$

where $f(x) = \frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} = \frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x+4)} = \frac{x-3}{x+4}$

$x \neq 1, 2, -4$

Range of $f(x) = R - \left\{1, \frac{-2}{5}, \frac{-1}{6}\right\}$

$\therefore$ $\frac{a}{30} \neq -1, \frac{2}{5}, \frac{1}{6}$

$\Rightarrow$ $a \neq -30, 12, 5$

8. (a), (b)

Here for the parabola $y^2 = 8x$, a = 2, Let $P(2t^2, 4t)$ be a point on parabola and $S(2, 0)$.

$SP = 4$

$\Rightarrow \quad a(1+t^2) = 4 \quad \Rightarrow \quad t = \pm 1$

P = (2, 4) or (2, – 4)

$f(x + y) = f(x)\, f(y)$; $f(1) = 2$; $f(2) = 2^2$

$f(n) = 2^n$; $a_1 = 2$; $a_2 = 4$

$\therefore \quad P \equiv (a_1, a_2)$ or $(a_1, -a_2)$.

9. (b), (d)

$$\int \frac{dx}{1-\sin^4 x} = \frac{1}{2}\left(\int \frac{dx}{1+\sin^2 x} + \int \frac{dx}{1-\sin^2 x}\right)$$

$$= \frac{1}{2}\int \frac{dx}{\cos^2 + 2\sin^2 x} + \frac{1}{2}\int \frac{dx}{\cos^2 x} = \frac{1}{2}\int \frac{\sec^2 x}{1+2\tan^2 x}dx + \frac{1}{2}\int \sec^2 x dx$$

$$= \frac{1}{2\sqrt{2}}\tan^{-1}(\sqrt{2}\tan x) + \frac{1}{2}\tan x + C\ .$$

10. (a), (b)

Since $\vec{a}$ and $\vec{b}$ are unit vectors

$\therefore \quad |\vec{a} - \vec{b}| = \sqrt{(\vec{a} - \vec{b})^2} = 2|\sin\theta| < 1$

$|\sin\theta| < \frac{1}{2}$; $\theta \in \left(0, \frac{\pi}{6}\right)$ or $\left(\frac{5\pi}{6}, \pi\right]$.

11. (a), (b), (c)

$\alpha^{11} = 1 \Rightarrow \alpha^{11} - 1 = 0 \Rightarrow (\alpha - 1)(\alpha^{10} + \alpha^9 + ... + \alpha + 1) = 0$

$\lambda = \alpha^6$ is a root of

$1 + x + x^2 + ...x^{10} = 0$ ..... (i)

$1 + \lambda + \lambda^2 + ... + \lambda^{10} = 0$

$\Rightarrow \quad 1 + \lambda + \lambda^2 + \lambda^3 + \lambda^4 + \lambda^5 + \lambda + \lambda^2 + \lambda^3 + \lambda^4 + \lambda^5 = 0$

$\text{Re}(\lambda + \lambda^2 + ...\lambda^5) = -\frac{1}{2}$

$\beta, \beta^2, \beta^3 .....\beta^{10}$ are roots of (i)

$\therefore \quad (x - \beta)(x - \beta^2)(x - \beta^3).....(x - \beta^{10}) = 1 + x + x^2 + ... + x^{10}$

Now put $x = \mu$

$(\mu - \beta)(\mu - \beta^2).....(\mu - \beta^{10}) = 0$ Put $x = i$

$(i - \beta)(i - \beta^2).....(i - \beta^{10}) = i$.

SECTION–III
Reasoning Type

12. **(d)** $f(x)=[x]^2-\sqrt{\{x\}^2}$ is continuous at $x = 1$.

13. **(b)** If each of x, y, z is less than $1/2$, then statement-1 is obviously true.

$\Rightarrow$ If one of x, y, z is more than or equal to $\frac{1}{2}$, then their product is less than equal to zero hence, statement-1 still remain true.

14. **(b)**

15. **(d)** $A = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$
Number of different rational numbers = $10p_2 + 1$,

(1 corresponding to numbers like $\frac{2}{2}$ or $\frac{3}{3}$)

$=10 \times 9 + 1 = 91$

16. **(d)** $\min\left(\sqrt{3},\ a^2-4a+5\right)\in[1,\ \sqrt{3}]$ and $(\sin x-2\cos x)\in[-\sqrt{5},\ \sqrt{5}]$
So there exist infinite solutions.

SECTION–IV
Matrix–Match Type

1. $f(x)>1 \Rightarrow x>0$
$f(x)<-1 \Rightarrow x<0$
$f(x)>3 \Rightarrow 0<x<\frac{1}{2}$
$f(x)<-3 \Rightarrow -\frac{1}{2}<x<0$

A-1; B-1, 3; C-2, 4; D-2

2. **A-1, 4, 5** Total number of cases $= 9!$
and number of favourable case are $= {}^9C_3 \times 6!$
Required probability $= \frac{{}^9C_3 \times 6!}{9!} = \frac{1}{6}$

B-2, 3 Required probability $= \frac{{}^9C_5 \times {}^4C_2 \times 4!}{9!} = \frac{1}{20}$

C-2, 3 Required probability $= \frac{{}^8C_5 \times {}^4C_2 \times 3!}{8!} = \frac{1}{20}$

D-1, 4, 5 Required probability $= \dfrac{{}^{9}C_3 \times {}^{6}C_3}{{}^{9}C_3 \times {}^{6}C_3 \times 3!} = \dfrac{1}{6}$

SECTION–V
Integer Answers Type

1. Solving the equation of the lines we get $z = -\bar{z} \Rightarrow z = i$

$|\alpha - i| = 2$; $\alpha = 2e^{i\theta} + i$, put it in the equation of the second line, we get

$\cos\theta - \sin\theta = 0$

$\Rightarrow \quad \theta = \dfrac{\pi}{4}$

$\alpha = i \pm 2e^{\frac{i\pi}{4}}$

$\therefore \quad x = \pm\sqrt{2}$

$\Rightarrow \quad [|x|] = 1$

Ans. 1

2. Let P be $(5\cos\theta, 4\sin\theta)$; Q be $(-5\sin\theta, 4\cos\theta)$

Equation of tangent at P $\dfrac{x}{5}\cos\theta + \dfrac{y}{4}\sin\theta = 1$ ..... (i)

Equation of tangent at Q $-\dfrac{x}{5}\sin\theta + \dfrac{y}{4}\cos\theta = 1$ ..... (ii)

Solving (i) and (ii) $\Rightarrow$ $R \equiv (5(\cos\theta - \sin\theta),\ 4(\sin\theta + \cos\theta))$

$\therefore \quad m : n$ is 1 : 1

$\Rightarrow \quad m + n = 2$

Alternate:

Let $P(5, 0)$; $Q(0, 4)$

$\Rightarrow \quad R(5, 4)$

Intersection of CR and PQ is $\left(\dfrac{5}{2}, 2\right)$, which is mid-point of CR

$\Rightarrow \ m:n = 1:1 \ \Rightarrow \ m + n = 2$

Ans. 2

3. $I = \int_0^1 \underset{\text{I}}{\left(1 - x^4\right)^7} \cdot \underset{\text{II}}{1}\, dx$

$= [x(1 - x^4)^7]_0^1 + 7 \times 4\int_0^1 x(1 - x^4)^6 x^3 dx$

$= -28\int_0^1 (1 - x^4)(1 - x^4)^6\, dx + 28\int_0^1 (1 - x^4)^6 dx = -28I + 28\int_0^1 (1 - x^4)^6 dx$

$29I = 28\int_0^1 (1 - x^4) dx$

$$\frac{29\int_0^1 (1-x^4)^7\,dx}{4\int_0^1 (1-x^4)^6\,dx} = 7$$

Ans. 7

4. $f'(x) = -a\pi\sin(\pi x)$

$\Rightarrow \quad f'\left(\frac{1}{2}\right) = -a\pi\sin\frac{\pi}{2} = -a\pi = \pi \;\Rightarrow\; a = -1$

$$\int (a\cos\pi x + b)dx = \left(\frac{a\sin\pi x}{\pi} + bx\right)_{1/2}^{3/2} = \left(\frac{-a}{\pi} + \frac{3b}{2}\right) - \left(\frac{a}{\pi} + \frac{b}{2}\right)$$

$$\Rightarrow \quad -\frac{2a}{\pi} + b = \frac{2}{\pi} + 1 \;\Rightarrow\; b = 1$$

So, $\frac{-12}{\pi}\left(\sin^{-1}(-1) + \cos^{-1}1\right) = \frac{-12}{\pi}\left(-\frac{\pi}{2} + 0\right) = 6$

Ans. 6

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